

# WILMS

Women's Interest-Free Loan Management System  
Operations Manual — v1.8.0

## WILMS Operations Manual

Version: 1.7.3 Audience: DevOps, Super Admins, programme IT leads Classification: Confidential

### 1. Purpose

This manual covers day-to-day operational procedures for running WILMS in production and staging environments.

### 2. Environment overview

| Environment      | URL                        | Purpose                   |
|------------------|----------------------------|---------------------------|
| ----- ---- ----- |                            |                           |
| Production       | wilms.vercel.app           | Live programme operations |
| Preview          | Vercel preview deployments | PR validation             |
| Local            | localhost:3000             | Development               |

### 3. Health monitoring

#### Health check

GET /api/wilms/health  
Returns service status, version, database connectivity.

#### Ops dashboard

Super Admin accesses /ops for:

- Active incidents
- Maintenance windows
- System metrics summary

### 4. Daily operations checklist

| Task                   | Frequency | Owner       |
|------------------------|-----------|-------------|
| ----- ----- -----      |           |             |
| Verify health endpoint | Daily     | DevOps      |
| Review ops incidents   | Daily     | Super Admin |

| Monitor reconciliation aging | Daily | Super Admin |

| Review notification cron logs | Daily | DevOps |

| Check Neon connection pool | Weekly | DevOps |

| Review audit log anomalies | Weekly | Auditor/Admin |

| Run smoke tests post-deploy | Per deploy | DevOps |

## 5. Deployment procedure

### Pre-deploy

1. Verify branch passes CI (lint, type-check, test). 2. Run `npm run verify:version`. 3. Run `npm run verify:migrations`. 4. Review CHANGELOG for release notes.

### Deploy

1. Merge to production branch. 2. Vercel auto-deploys on push. 3. Migrations applied: `npm run db:migrate -w @wilms/domain`. 4. Post-deploy smoke: `npm run smoke:production`.

### Rollback

1. Revert Vercel deployment to previous build. 2. Database rollback requires manual SQL — migrations are forward-only. 3. Document incident in ops module.

## 6. Database operations

### Connection

Neon serverless PostgreSQL via DATABASE\_URL. Connection pooling enabled.

### Migrations

`npm run db:migrate -w @wilms/domain npm run verify:migrations -w @wilms/api`

### Backup

Neon provides automated point-in-time recovery. Verify backup retention in Neon dashboard. Run restore drill: `npm run drill:backup-restore`.

### Demo data cleanup

`npm run cleanup:demo-financial-data -w @wilms/domain`

### Location master import

`npm run seed:location-master -w @wilms/domain npm run db:backfill:locations -w @wilms/domain`

### Reset transactional data (keep users and RBAC)

`WILMS_CONFIRM_DB_RESET=YES npm run db:reset:keep-users -w @wilms/domain`

Preserves users, roles, permissions, role\_permissions, user\_roles, user\_permission\_overrides, and \_\_drizzle\_migrations. Then re-import locations.

## 7. Cron and scheduler

| Job | Schedule | Endpoint |

|----|-----|-----|

| Notification dispatch | 06:00 UTC daily | /api/cron/notifications |

Configured in vercel.json. Verify cron execution in Vercel dashboard logs.

On each run the payment scheduler sends, once per qualifying week (deduped):

| Borrower message | Timing |

|-----|-----|

| Payment reminder | One day before due date (paymentReminderDaysBefore, default 1) |

| Due today | Morning of payment day (06:00 UTC cron) |

| Missed payment | After the due date when the week is marked missed |

| Grace reminder | On the last grace day (latePaymentGraceDays) |

| Escalation | The day after grace expires |

Admin-fee instruction is sent at loan approval, not at registration. Disbursement is blocked until the admin fee is recorded. Full copy and channel matrix: [documentation/notifications/](#).

## 8. User management operations

### Invite new user

Super Admin ! Settings ! Users ! Invite. User receives email with invitation link INVITED to ACTIVE.

### Force logout

Super Admin ! User profile ! Force logout. Invalidates all active sessions for

### Permission overrides

Grant individual permissions beyond role defaults. All overrides audited.

### Password reset

Users can self-service via forgot-password flow. Admin cannot view passwords.

## 9. Incident response

### Severity levels

| Level | Description | Response time |

|-----|-----|-----|

| P1 | Production down, data integrity risk | Immediate |

| P2 | Major feature degraded | 4 hours |

| P3 | Minor issue, workaround available | 24 hours |

| P4 | Cosmetic, documentation | Next sprint |

## P1 procedure

1. Create ops incident in WILMS. 2. Check health endpoint and Vercel status. 3. Check Neon status page. 4. Rollback deployment if recent deploy caused issue. 5. Notify programme director. 6. Post-incident review within 48 hours.

## 10. Maintenance windows

Schedule via Ops module before:

- Database migrations with downtime risk
- Major version deployments
- Neon maintenance periods

Maintenance window displays banner to all users.

## 11. Security operations

| Task | Procedure |

|-----|-----|

| Rotate session secret | Update WILMS\_SESSION\_SECRET in Vercel; all users re-login |

| Rotate production users | npm run rotate-production-users -w @wilms/domain |

| Review failed login attempts | Audit log filter by auth.login\_failed |

| Upload audit | Review upload logs for anomalous file types |

## 12. Performance monitoring

- Bundle budget: npm run bundle:budget-check
- Performance budget: npm run perf:budget-check
- API rate limit: 300 req/min global; monitor 429 responses

### 12a. Ghana location hierarchy

Refresh official geography after a licensed dataset update:

npm run db:apply:ghana-hierarchy -w @wilms/domain npm run seed:ghana-hierarchy -w @wilms/domain

Confirm GET /api/v1/locations/sync/status. Do not delete historical location rows. Community suggestions are Super Admin review items, never auto-created.

## 13. Documentation maintenance

Generate updated PDF/DOCX after documentation changes:

```
npm run docs:generate
```

Outputs in documentation/pdf/ and documentation/docx/.

## 14. v1.8.0 market-readiness operations

- Apply migration 0044\_v180\_borrower\_update\_requests.sql before using borrower update requests.
- Review Pending borrower update requests daily (/officer/borrower-updates or /borrower-updates).
- Migration 0044 (borrower\_update\_requests) is applied in production — see documentation/release/MIGRATION\_0044\_VERIFICATION.md.
- Final production release evidence:  
documentation/release/WILMS\_v1.8.0\_FINAL\_PRODUCTION\_RELEASE\_REPORT.md.
- If field GPS fails, collectors must record a reason; audit action collection.gps-exception.
- Disable SMS only for a confirmed provider outage (smsNotificationsEnabled).

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